Austin Brown | C.V.

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Research interests

My main research focus is on the design, reliability, and computational efficiency of Markov chain Monte Carlo algorithms by studying their convergence properties. Creating reliable, computationally efficient Markov chain Monte Carlo algorithms for practitioners is a central motivation for my research. Additionally, I am interested in applications of Markov chain Monte Carlo to Bayesian error-in-variable models used in astrophysics and epidemiology. My other research interests are optimal transportation and weak convergence of stochastic processes.

My dissertation "Some Convergence Results for Metropolis-Hastings Algorithms" is available online `https://conservancy.umn.edu/handle/11299/243073`.

Education

University of Minnesota - Twin Cities **Minneapolis, MN**
Ph.D. in Statistics *August 2022*
Dissertation: "Some Convergence Results for Metropolis-Hastings Algorithms"
Advisor: Galin L. Jones
Committee: Charles Geyer, Qian Qin, Wei-Kuo Chen

University of Florida **Gainesville, FL**
B.S. in Mathematics *July 2017*
Major in Mathematics and minor in Statistics

Appointments

University of Toronto - St George **Toronto, Ontario**
Postdoctoral Fellow *July 2023–July 2025*
Advisor: Jeffrey Rosenthal

University of Warwick **Coventry, England**
Research Fellow *September 2022–June 2023*
Advisor: Krzysztof Łatuszynski

Publications and preprints

[1] A. Brown and J. S. Rosenthal, "Upper and lower bounds on the convergence of adaptive Markov chain Monte Carlo," *In progress*, 2024.

[2] A. Brown and J. S. Rosenthal, "Weak convergence of adaptive Markov chain Monte Carlo," *minor revision requested, Journal of Applied Probability*, 2024.

[3] S. Sixta, J. S. Rosenthal, and A. Brown, "Bounding and estimating MCMC convergence rates using common random number simulations," *ArXiv preprint, in peer review Stochastic Models*, 2024.

[4] A. Brown, "A non-asymptotic error analysis for parallel Monte Carlo estimation from many short Markov chains," *ArXiv preprint, In peer review at Statistics and Computing*, 2024.

[5] A. Brown, K. Łatuszyński, and G. Roberts, "Adaptive pseudo-marginal Metropolis-Hastings," *In progress*, 2023.

[6] A. Brown and G. L. Jones, "Exact convergence analysis for Metropolis–Hastings independence samplers in Wasserstein distances," *Journal of Applied Probability*, vol. 61, no. 1, p. 33–54, 2024.

[7] A. Brown and G. L. Jones, "Lower bounds on the rate of convergence for accept-reject-based Markov chains in Wasserstein and total variation distances," *Bernoulli, to appear*, 2024.

[8] A. Brown, "Geometric ergodicity of Gibbs samplers for Bayesian error-in-variable regression," *Electronic Journal of Statistics*, vol. 18, no. 1, pp. 1495–1516, 2024.

[9] A. Brown and G. L. Jones, "Convergence rates of Metropolis–Hastings algorithms," *WIREs Computational Statistics*, vol. 16, no. 5, p. e70002, 2024.

Seminar talks and conference posters

University of Warwick **Coventry, United Kingdom**
Algorithms and Computationally Intensive Inference Seminar *January 2023*
Talk Title: Lower Bounds on the Rate of Convergence for Accept-Reject-Based Markov Chains
Slides: `https://austindavidbrown.github.io/talk/warwick2023.pdf`

University of Warwick **Wales, United Kingdom**
University of Warwick Departmental Conference 2023 *April 2023*
Talk Title: Exact convergence for independence samplers in Wasserstein distance.
Slides: `https://austindavidbrown.github.io/talk/warwickdepartmental2023.pdf`

University of Oxford **England, United Kingdom**
Bioinference 2023 *June 2023*
Poster Title: Geometric ergodicity of Gibbs samplers for Bayesian error-in-variable regression
Poster: `https://austindavidbrown.github.io/poster/bioinference2023.pdf`

University of Florida **Gainesville, FL**
University of Florida Statistics Seminar *September 2023*
Talk Title: Lower Bounds for Metropolis-Hastings in Wasserstein distances
Slides: `https://austindavidbrown.github.io/talk/uf2023.pdf`

University of Toronto **Toronto, Canada**
University of Toronto B.B. Seminar *January 2024*
Talk Title: Error analysis for parallel Monte Carlo estimation from many short Markov chains
Slides: `https://austindavidbrown.github.io/talk/toronto2024.pdf`

Teaching experience

University of Minnesota - Twin Cities **Minneapolis, MN**
Instructor ($\sim$ 70 students) *Spring 2020*
Regression and Correlated Data (STAT 3032)

University of Minnesota - Twin Cities **Minneapolis, MN**
Instructor ($\sim$ 80 students) *Spring 2021*
Introduction to Statistical Analysis (STAT 3011)

University of Toronto - St. George **Toronto, Canada**
Instructor ($\sim$ 250 students) *Fall 2023*
Methods of Data Analysis 1 (STAT 302)

University of Toronto - St. George **Toronto, Canada**
Instructor ($\sim$ 300 students) *Fall 2024*
Methods of Data Analysis 1 (STAT 302)

Academic service

I have referred 6 articles for statistics and applied probability journals including Bayesian Analysis, Journal of Computational and Graphical Statistics, PLOS ONE, Advances in Applied Probability, Journal of Applied Probability, and Sankhya, Series B. I have also assisted in editing *An Introduction to Envelopes* by Dennis Cook.

I have taken part in the following committees and organizations:

American Statistical Association LGBTQ+ Advocacy Committee *August 2023–Present*

Organizer of University of Warwick Algorithms and Computational Intensive Seminar *Jan 2023–June 2023*

University of Warwick Statistics Department IT Committee *2022–June 2023*

University of Warwick Stonewall Self-Assessment Team (LGBTQ+ Inclusion) *2022–June 2023*

Co-organizer of University of Toronto Brown Bag Departmental Seminar *September 2023–Present*

Organizations and memberships

Institute of Mathematical Statistics *2022–Present*

Funding, scholarships, and awards

Beverly and Richard Fink Fellowship ($56000+Tuition) *2017–2022*

Lynn Lin Fellowship ($2000) *2019–2019*

First Year Scholarship ($2500) *2017–2017*

Mentorships

University of Toronto - St George **Toronto, Canada**

2024

I have acted as a mentor to 2 PhD students and 1 undergraduate student on their own individual research projects

Extracurricular activities

- Markov Chain Monte Carlo Group (2019–2022)
  Discussion of the state of the art MCMC literature

Statistical software

cmhi: A Python package for the centered Metropolis-Hastings independence algorithm. github.com/austindavidbrown/Centered-Metropolis-Hastings

mhlb: A Python implementation to estimate lower bounds on the geometric convergence rate for RWM Metropolis-Hastings. https://github.com/austindavidbrown/lower-bounds-for-Metropolis-Hastings

Programming language proficiency

Python: Pytorch, Autograd, Numpy, Matplotlib *Expert*

R: Rmarkdown *Expert*

C++: C++14 specification and above *Intermediate*

Languages

- English (native)
- French (novice)
- Spanish (novice)

References

- Galin Jones. University of Minnesota - Twin Cities. Email: galin@umn.edu
- Qian Qin. University of Minnesota - Twin Cities. Email: qqin@umn.edu
- Jeffrey S. Rosenthal. University of Toronto. Email: jeff@math.toronto.edu
- Krzysztof Latuszynski. University of Warwick. Email: K.G.Latuszynski@warwick.ac.uk